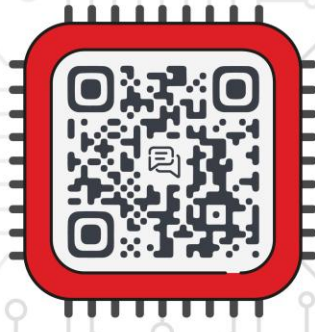




نظري

- كتاب
66
7920 sp

كلية الهندسة المعلوماتية

السنة الثالثة

ملحق امتحاني

عمل طلابي



محتوى مجاني غير مخصص للبيع التجاري
12/02/2024
بنيان الحواسيب 2

أهلاً بكم في ملحق مادة بنيان الحواسيب 2

يحتوي هذا الملحق على تلخيص للمادة وأسئلة دورات ومواقع ، وننوه انه من الأفضل دراسة المحاضرات أولاً و نظراً لأن الأسئلة في الامتحان ستكون باللغة الإنجليزية فقد تمت كتابة بعض الشروحات والمصطلحات بالإنجليزية لضمان استيعابها بشكل أفضل .

ونرجو من الله الفائدة و التوفيق لكم ولنا 😊

إلى الصفحة	من الصفحة	
13	2	ملخص لبعض الأفكار
49	14	أسئلة قسم الدكتوراة خولة
66	50	أسئلة قسم الدكتوراة سيرا



هيا بنا ننتقل...

Performance:

- RISC (Reduced Instruction Set Computer) $\Rightarrow$ hardware
- CISC (Complex Instruction Set Computer) $\Rightarrow$ software

Computer architecture (software):

Refers to those attributes of a system visible to a have a direct programmer or, put another way, those attributes that impact on the logical execution of a program.

Computer organization (hardware):

Refers to the operational units and their interconnections that realize the architectural specifications.

Organizational attributes include those hardware details transparent to the programmer, such as control signals; interfaces between the computer and peripherals; and the memory technology used.

يعتمد الأداء على:

Compiler II

لغة البرمجة

الخوارزمية المستخدمة

Hardware II

بنية البرنامج

تعريف:

- **Response time:** The time between the start and completion of a task.
- **Execution time:** The total time required for the access computer to complete a task, including: the disk memory, I/O activities, Operating system overhead, cpu execution time.
- **Latency:** How long to finish my program.
- **Throughput:** How much work finished per unit time.

$$CPI = \frac{CC}{IC} \quad \begin{array}{l} \text{(عدد الدورات خلال التعليم)} \\ \text{عدد دورات البرنامج / عدد التعليمات} \end{array}$$

$$IPC = \frac{1}{CPI}$$

$$\begin{aligned} CPU \text{ time} &= \text{Instruction count} \times CPI \times \text{Clock cycle time} \\ &= CC \times \text{clock cycle time} = \frac{CC}{\text{clock rate}} \end{aligned}$$

$$\text{Clock rate (Hz)} = \frac{1}{\text{clock cycle time}}$$

$$\text{Performane} = \frac{1}{\text{execution time}}$$

(تناسب عكسي بين الأداء و زمن التنفيذ/زمن معالجة كل التعليمات)

$$CPU \text{ execution time for a program} = \left(\sum_{i=1}^n (CPI_i \times I_i) \right) \times \text{clock cycle time}$$

عدد دورات التعليم
عدد التعليمات من النوع i

$$MIPS = \frac{\text{Instruction count}}{\text{execution time} \times 10^6} = \text{clock rate} \frac{1}{CPI \times 10^6}$$

إذا كان هناك تعليمات مختلفة تأخذ عدد مختلف من الدورات:

$$\text{clock cycles} = \sum_{i=1}^n (CPI_i \times \text{Instruction count}_i)$$

لحساب الـ CPI الوسطي:

$$CPI = \frac{\text{clock cycle}}{\text{Instruction count}} = \sum_{i=1}^n \left(CPI_i \times \frac{\text{Instruction count}_i}{\text{Instruction count}} \right)$$

Amdahl's Law:

Execution time after improvement =

$$\left(\frac{\text{execution time affected by improvement}}{\text{amount of improvement}} \right) + \text{execution time unaffected}$$

عند حساب القيم يجب مراعاة الواحدات المتقابلة كالتالي :

Clock speed (هرتز) أو دورة في الثانية) Hz)

- 1 KHz = 10^3 Hz : cycle in 1 millisecond (10^{-3} second)
- 1 MHz = 10^6 Hz : cycle in 1 microsecond (10^{-6} second)
- 1 GHz = 10^9 Hz : cycle in 1 nanosecond (10^{-9} second)
- 1 THz = 10^{12} Hz : cycle in 1 picosecond (10^{-12} second)

Multi cycle instruction:	Single cycle instruction:
تكون قيمة CPI فيه مختلفة حسب كل تعليمة.	تكون قيمة CPI فيه تساوي الواحد أيًا.

Control unit Operation :

كل تعليمة عند تنفيذها تمر بعدة مراحل :

Fetch (3) :

- $t_1: \text{MAR} \leftarrow (\text{PC})$
- $t_2: \text{MBR} \leftarrow (\text{memory})$
- $\text{PC} \leftarrow (\text{PC}) + 1$
- $t_3: \text{IR} \leftarrow (\text{MBR})$

نلاحظ ان هذه العملية تحتاج 3 دورات .
والتعليمتين الصغريتين الأخيرتين يمكن ان يتما سوياً في الدورة الثالثة .

العنونة غير المباشرة (indirect address) :

- $\text{MAR} \leftarrow (\text{IR}_{\text{address}})$
- $\text{MBR} \leftarrow (\text{memory})$
- $\text{IR}_{\text{address}} \leftarrow (\text{MBR address})$

دورة المقاطعة (Interrupt cycle) :

- $t_1: \text{MBR} \leftarrow (\text{PC})$
- $t_2: \text{MAR} \leftarrow \text{save-address}$
- $\text{PC} \leftarrow \text{routine-address}$
- $t_3: \text{memory} \leftarrow (\text{MBR})$

دورة التنفيذ (Execution cycle) :

تختلف حسب نوع التعليمات ، وفي ما يلي ما تم ذكره في المحاضرة :

1. ADD R1, X:

- $t1: MAR \leftarrow (IR_{address})$
- $t2: MBR \leftarrow (memory)$
- $t3: R1 \leftarrow R1 + (MBR)$

2. ISZ (increment and skip if zero) :

- $t1: MAR \leftarrow (IR_{address})$
- $t2: MBR \leftarrow (memory)$
- $t3: MBR \leftarrow (MBR) + 1$
- $t4: memory \leftarrow (MBR)$
- if $(MBR) == 0$ then $PC \leftarrow (PC) + 1$

انتقل للتعليمات التي تليها → نتيجة الإضافة كانت صفر

3. BSA (Branch and save address) :

- $t1: MAR \leftarrow (IR_{address})$
- $MBR \leftarrow (PC)$
- $t2: PC \leftarrow (IR_{address})$
- $memory \leftarrow (MBR)$
- $t3: PC \leftarrow (PC) + 1$

كل عملية في المعالج تنقسم الى عدة تعليمات صغيرة وعند ترميز التعليمات الصغيرة يتم تخصيص كود لكل تعليمات وتسمى :

■ ICC: instruction cycle code

فمثلاً نحن هنا لدينا أربع مراحل صغيرة للتنفيذ نريد ترميزها وترمز باستخدام 2 بت :

■ حسب $2^{\text{عدد البتات}} = \text{عدد المراحل}$

00: Fetch 01: Indirect 10: Execute 11: Interrupt

وظائف وحدة التحكم :

1. المحافظة على التسلسل (sequencing).
2. التنفيذ (Execution) .
3. تقوم وحدة التحكم بعملها من خلال إشارات التحكم التي تولدها (control signal) .

■ دخل وحدة التحكم (Control Signals-Input) :

1. clock

3. Flags

2. instruction register (IR)

4. Interrupts, Acknowledgements

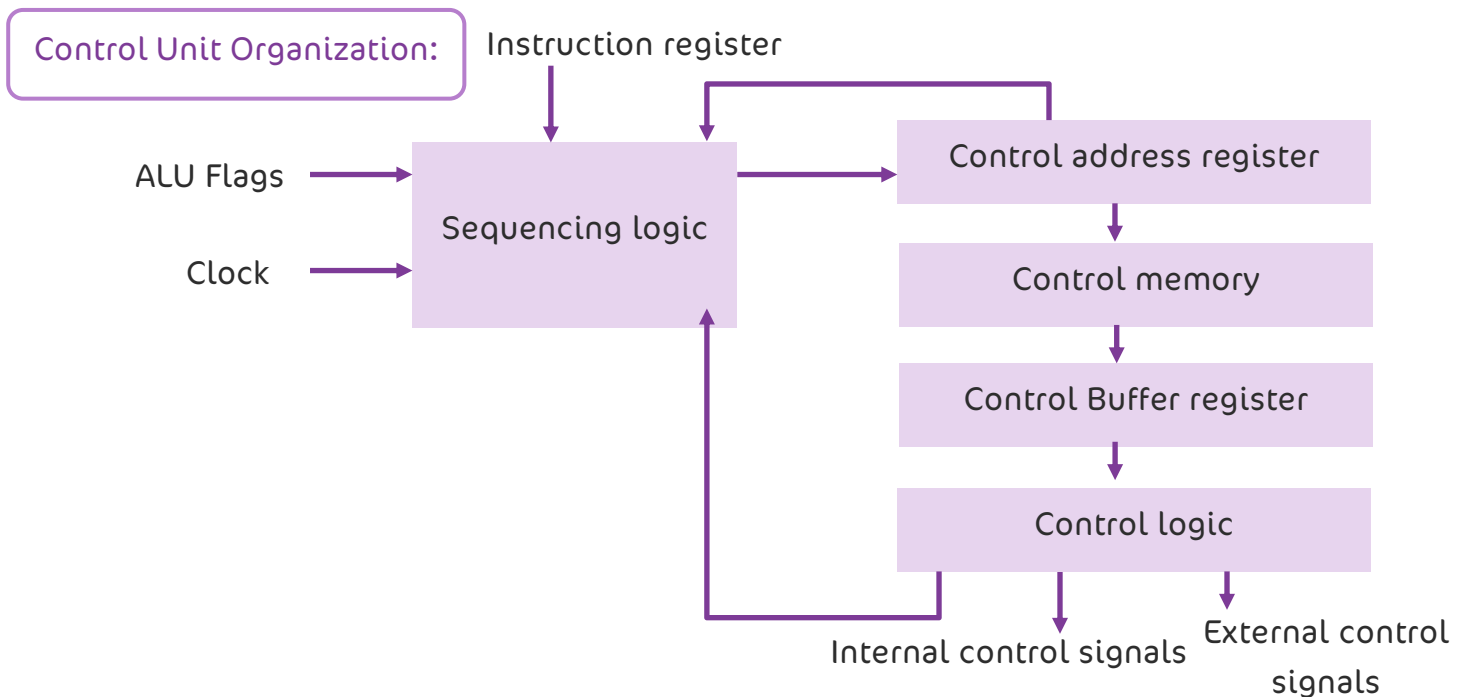
يحتوي على (op-code) التعليمات التي ستنفذ

■ خرج وحدة التحكم (Control Signals-output) :

2. (عبر ناقل التحكم) via control Bus :

1. (إشارات تحكم ضمن CPU) within CPU :

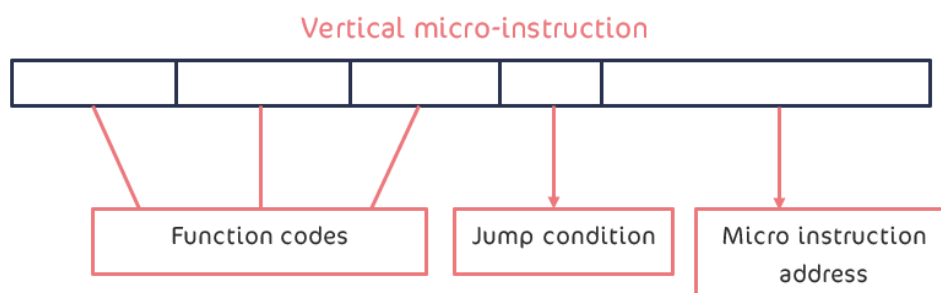
- تسبب تحريك البيانات.
- تفعل وظائف خاصة أو محددة.
- مثلاً: تنفيذ عملية جمع، طرح، قسمة، عملية منطقية، القراءة و الكتابة من و على السجلات....
- الذاكرة (تخزين - تحميل).
- مجتزأت الدخل و الخرج.



أنواع التعليمات الصغرية (Micro Instruction Types) :

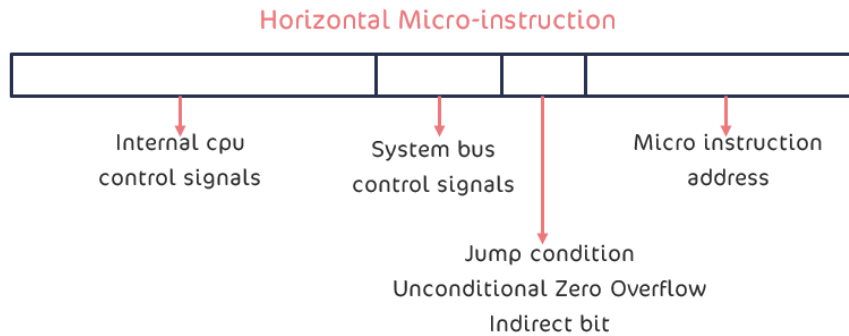
1-vertical micro operation :

- -Width is narrow
- -n control signals encoded into $\log_2 n$ bits
- -Limited ability to express parallelism
- -Considerable encoding of control information requires external memory word decoder to identify the exact control line being manipulated .



2-horizontal micro operation:

- Wide memory word
- High degree of parallel operations possible
- Little encoding of control information



Control Unit Design :

Data path and control :

1. **Data path**: هو الطريق الذي تسلكه البيانات وهذا الطريق يمر بالعديد من قطع الحاسوب . حيث أنه كل خطوة في معالجة عملية ما تتطلب نقل بيانات (أي اتصال) و تتطلب ممرات بين الذاكر والسجلات و وحدة الحساب و المنطق .
2. **Control**: إن ممر التعليم هو نظام تم فرضه من قبل إشارات التحكم (كل تعليمة لها ممر خاص) حيث تنظم هذه الإشارات اتجاهات تدفق البيانات في قنوات الاتصال و تختار الوظائف المناسبة لكل من الذاكرة و وحدة الحساب و المنطق .

There are two design methods

Bottom-up composition of primitive building blocks into more complex assemblies .

Top Down of complex function behavior into more primitive functions

ملاحظات :

الدارة المتتابعة (sequential) لا تمثل بجدول حقيقة و إنما تمثل ب FSM .

FSM: تتألف من مجموعة حالات واتجاهات تبين تغير الحالة ، فنحتاج في تصميمها إلى Flipflops أي إذا كان لدينا n bits تخزين (n قلاب) فيمكن تمثيل n^2 حالة على الأكثر .

يحدد تابع الحالة التالية logic (**next state func**) أي منطقياً باستخدام ROM و PLA و هي عبارة عن شرائح فيها العديد من البوابات المنطقية يتم وصلها لتشكيل دارة تركيبية PLA .

هناك طريقة بديلة لتنفيذ هذا التصميم حيث أنه يتم التحكم بتابع الحالة التالية باستخدام **Counter** عداد .

FSM Gray code Example

- | | |
|----------------|--|
| 1) First step | state Diagram |
| 2) Second step | next state table |
| 3) Third step | Flip-Flop Transition Table |
| 4) Forth step | Karnaugh tables |
| 5) Fifth step | Logical Expressions for flip-flops inputs. |
| 6) Sixth step | counter implementation |

هناك خمس أنواع عامة للتعليمات

- 1) 3-operand ALU
- 2) ALU w/immediate
- 3) Loads/Stores
- 4) Branches
- 5) Jumps

تصميم وحدة التحكم الرئيسية (Designing Main Control Unit) :

- تصميم وحدة التحكم الرئيسية متعدد المستويات، حيث أن المستوى الأساسي هو main control unit
- يخرج منها مستويين يعطيان خطوط تحكم، هذا التصميم لـ Single cycle.
- سوف نمتلك وحدة تحكم لـ ALU بالإضافة لوحدة التحكم الرئيسية.

ملاحظات عن بنية تعليمات MIPS :

- حقل الـ Opcode دائماً ثابت الموقع والحجم حيث موقعه البتات $26 \leftarrow 31$ و هي البتات الستة الأخيرة اليسارية.
- $rs : 21 \leftarrow 25$ بت ; $rt : 16 \leftarrow 20$ بت
- و هذا ينطبق على التعليمات من النوع R-Type, branch equal, store Instructions.
- السجل الأساسي للتخزين والتحميل هو rs .
- الـ 16 بت المخصصين لتعليمات store-load-branch يدلون على العنوان موقعهم في البتات $0 \leftarrow 15$
- سجل الوجهة يكون إحدى المكانين:

للتحميل (تعليمة load) تكون في السجل rt الذي موقعه البتات $16 \leftarrow 20$.

التعليمات من نوع R-Type في السجل rd الذي موقعه البتات $11 \leftarrow 15$ نحتاج ناخب (Mux) لاختيار حقل التعليمة المستخدم.



في multi-cycle لا يوجد سوا ذاكرة واحدة فقط ولم نحتاج ل ALU ثانية لزيادة عداد PC مثل Single cycle تصميم ال

- R-type : 4 cycle
- load= 5 cycle , store = 4 cycle
- Branch + J-Type => 3 cycles .

Memory Hierarchy & Cache memory :

مبدأ التقارب (principle of locality) :

1. تقارب زمني (Temporal locality) :

هناك تعليمات تم الوصول لها مؤخراً يمكن أن يُعاد استخدامها مرة أخرى في وقت قريب (loops) .

2. تقارب مكاني (Spatial locality) :

تعليمات متتالية سيتم تنفيذها تتابعياً ، (arrays) .

الذاكر (Memory) :

Speed and cost تزداد السرعة و الكلفة ↓	Disk ⇒ store everything .
	DRAM ⇒ copy recently accessed item to it from disk . يتم نسخ آخر شيء تم الوصول إليه من القرص إلى DRAM .
Capacity and access time تزداد السعة و زمن الوصول ↑	SRAM ⇒ copy more recently accessed item from DRAM to it . ثم يتم النسخ من DRAM إلى SRAM .

عند تنفيذ أي أمر برمجي أو غير برمجي يحتاج الوصول لـ data في الذاكرة ، فهذا الشيء يأخذ وقت فظهرت الـ cache كذاكرة مؤقتة أقرب لـ CPU من الـ main memory حيث يتم نسخ المعلومات التي تحتاجها إليها .

Ex : cache memory , main memory .

Block : unit of data transfer between memories .

يتم نسخ (الداتا بين مستويين متتاليين ، (n , n+1) وليس (n , n+2) / حيث تعبر الأرقام عن (المستوى)

- عندما يتم طلب معلومة من ذاكرة عالية المستوى (الكاش مثلاً)

حالة Hit

الداتا موجودة في المستوى الأعلى (Cache)

$$Hit\ ratio = \frac{hits}{accesses}$$

حالة Miss

الداتا غير موجودة في الكاش فيتم نسخها من المستوى الذي قبلها (RAM)

$$Miss\ ratio = \frac{misses}{access} = 1 - Hit\ ratio$$

$$Miss\ ratio + Hit\ ratio = 1$$

DRAM	Flash Storage
• هي Dynamic لأنها تتجدد بشكل دوري (periodically refreshed)	• Non volatile semiconductor storage
• الداتا هنا تمر في أسطر و أعمدة و التي تخزن ضمن Bank	• أسرع من القرص 100 ← 1000 مرة .
• Data stored as a charge in a capacitor	

Disk storage	Cache memory
• Consists of a collection of platters	• A small fast memory that acts as a buffer
• Each surface divides into tracks	• It is closet to the CPU
• Each track → sectors	• نحمل مجموعة التعليمات ذات الاستخدام المتكرر .

أنواع ذاكرة Cache حسب العلاقة بين عنوان البيانات في الذاكرة الرئيسية و موقعها في (Mapping) cache

Direct mapped cache

- Sets : Blocks
- حيث كل مجموعة تحتوي block فقط .
- كل عنوان في الذاكرة الرئيسية يمكن أن يتموضع في مكان واحد فقط في الكاش

N-way set associative cache

- Set $\rightarrow$ N-Block
- كل مجموعة فيها Block N
- كل عنوان يمكن أن يخزن في مجموعة واحدة فقط (في أي بلوك موجود في المجموعة) .

Fully associative cache

- مجموعة واحدة و هي كل الكاش أي أن ممكن للعنوان أن يتوضع في أي Block

يحدد موقع Block في الـ cache حسب القانون:

$$\frac{\text{عنوان الـ Block في الذاكرة}}{\text{عدد الـ Blocks في الـ cache}} = \text{باقي قسمة}$$

Cache example :

- مثلاً لدينا عنوان ذاكرة نريد البحث عنه مؤلف من عدد بتات ، يكون الـ n بت على اليمين (حيث n عدد بتات تمثيل محل Block في الكاش) تمثل عنوان الداتا في الكاش و تسمى index .
- البتات العليا الباقية تميز بين العناوين في الذاكرة ذات الـ index (عناوين الكاش) المشابهة ، و تسمى Tags .
- نضيف بت أعلى يسمى valid bit يبين فيما إذا كانت الداتا (العنوان) موجود في الـ cache أم لا .
- (0 غير موجود miss) ، (1 موجود hit) .

Address Subdivision :

1

Total number of bits in cache: $= 2^n \times (\text{block size} + \text{tag size} + \text{valid field size})$
 block size: عدد بتات البلوك الواحد ; valid field size = دائماً 1

2

n : عدد بتات الـ index ; حيث 2^n سعة ذاكرة الكاش من الـ Blocks .

3

Tag size = block size - (n+m+2)

4

كل block هو 2^m كلمة (word) أو 2^{m+2} bytes ; و من هنا نحسب m
 (عدد بايتات كل بلوك) $m+2 = \log_2$; (عدد كلمات كل بلوك) $m = \log_2$

Associative Caches :

Associative caches

Fully Associative

نحتاج لعدد مقارنات مساوي لعدد أسطر المجموعة
و هنا هي كل الكاش أي مساوي لعدد كل
ال blocks .

n-way set associative

نحتاج لعدد مقارنات مساوي لعدد أسطر (blocks)
كل مجموعة (n) .
حيث أن كل مجموعة تحوي n سطرًا .

- أما في direct mapped نحتاج مقارن واحد حيث أن كل مجموعة هي Block واحد فقط .
- يتم التقسيم المجموعات بحيث كل مجموعة تحوي على عدد من مضاعفات ال 2 من الأسطر .

إذا كنا نريد التعديل على Block معين من
الكاش أي أن المعالج يريد الكتابة على الذاكرة

ذاكرة الكاش لا تحوي ال Block المطلوب

Write miss

فإما يحصل :

- Write allocate
- No write allocate

ذاكرة ال cache تحوي ال Block المطلوب

Write hit

فإما يحصل :

- Write through
- Write Buffer
- Write back

$$\text{Memory Stall Cycles} = \text{Memory accesses per program} \times \text{miss rate} \times \text{miss penalty}$$

$$\text{Average Memory Access Time (AMAT)} = \text{Hit Time} + (\text{Miss rate} * \text{Miss Penalty})$$

Virtual machine and memory :

- Virtual memory : هي تقنية تستخدم جزءاً من القرص الصلب كامتداد لذاكرة RAM عندما تكون غير كافية لتحميل البرنامج الذي يتم تنفيذه .
- Virtual machine : هي بيئات محاكاة تسمح بتقسيم الحاسوب إلى عدة حواسيب افتراضية ، كل منها يمكن أن يعمل بنظام تشغيل مختلف .

Virtual machine monitor : إدارة الآلات و ربطها بالداتا التي تحتاجها كدخل و خرج .
 Instruction set support : دعم مجموعة التعليقات المتعلقة بأنظمة التشغيل الافتراضي .
 عند تشغيل الآلات الافتراضية يكون لدينا وضعين :

System mode

يمكن الوصول لكل الموارد هنا و ذلك باستخدام
 privileged instruction لا

User mode

تكون العمليات التي يمكن القيام بها محدودة
 و لا يمكن الوصول للداتا و الموارد مباشرة

- إذا حاول برنامج ما في user mode استخدام privileged instruction يتم عمل Trap إلى System mode لمعالجة الطلب وذلك لحماية النظام من الاستخدام غير المصرح به .
- الموارد التي يحتاجها البرنامج ممكن أن تكون داتا أو :
 (page table , interrupt controls , I/O Registers)

العناوين هنا نوعين :

فيزيائية (فعلية)

v.page number | offset

افتراضية

ph.page number | offset

يحدث Page fault عند محاولة الوصول إلى صفحة غير موجودة في ذاكرة الوصول العشوائي RAM لأنه تم تخزينها في القرص الثانوي .

آليات إدارة الذاكرة :

Disk write

Page replacement

الهدف تقليل حدوث أخطاء الصفحات باستخدام
 خوارزمية استبدال الأقل استخداماً مؤخراً LRU

TLB (Translate Look – aside Buffer) : تخزين نسخ من مدخلات جداول الصفحات ،
 فتسرع عملية الوصول للداتا ، وتتعامل مع حالة page fault .

قسم الدكتور خولة

1- Which of the following best defines computer architecture?

- A. The physical arrangement of hardware components
- B. The attributes of a system visible to a programmer
- C. The way control signals are processed inside the CPU
- D. The interaction between peripherals and the computer

الإجابة : B

2- Computer architecture affects which of the following?

- A. The way a program is logically executed
- B. The placement of components inside the CPU
- C. The method used to manufacture a processor
- D. The arrangement of circuit boards inside a computer

الإجابة : A

3- Which of the following is an example of a computer architecture attribute?

- A. Type of control signals in a processor
- B. The number of bits used to represent characters
- C. The wiring connections inside a motherboard
- D. The cooling mechanism of a processor

الإجابة : B

4- What does computer organization focus on?

- A. The operational units and their interconnections
- B. The user interface of the operating system
- C. The programming languages used in software development
- D. The number of registers in a programming language

الإجابة : A

5- Which of the following is a characteristic of computer organization?

- A. I/O mechanisms
- B. Memory addressing techniques
- C. Control signals and interfaces between components
- D. The size of the cache memory

الإجابة : C

6- Which of the following would be considered part of computer architecture rather than organization?

- A. The number of CPU cores in a system
- B. The technique used for virtual memory management
- C. The arrangement of ALU and registers in a CPU
- D. The electrical wiring of data buses

الإجابة : B

7- What is the primary difference between computer architecture and computer organization?

- A. Computer architecture defines how a system is physically built, while computer organization defines programming languages.
- B. Computer organization focuses on system interconnections, while computer architecture focuses on logical execution
- C. Computer architecture deals with power consumption, while computer organization deals with security
- D. Computer architecture is related to the cooling system of a computer, while computer organization is about the user interface.

الإجابة : B

8- What is the function of a quartz crystal oscillator in a computer?

- A. To generate digital clock pulses directly
- B. To generate an analog pulse for timing signals
- C. To store digital data temporarily
- D. To perform arithmetic operations

الإجابة : B

9- Why does the clock circuit need to convert the pulse from the quartz crystal oscillator?

- A. Because the computer cannot process analog signals directly
- B. Because the clock circuit does not support quartz oscillators
- C. Because analog signals are slower than digital signals
- D. Because digital signals consume more power

الإجابة : A

10- Why is a clock pulse important in a computer?

- A. It keeps track of time for the operating system
- B. It synchronizes the operations of the processor and components
- C. It generates power for the CPU
- D. It converts data from binary to decimal

الإجابة : B

11- What is the primary function of the Control Unit (CU) in a CPU?

- A. Storing frequently used instructions
- B. Controlling the execution of instructions by directing data flow
- C. Performing calculations and comparisons
- D. Managing power supply to the CPU

الإجابة : B

12- What is the purpose of CPU registers?

- A. To permanently store data
- B. To temporarily store data and instructions during processing
- C. To increase the speed of the internet
- D. To store all installed applications

الإجابة : B

13- Which of the following registers stores the instruction that the CPU is currently executing?

- A. Program Counter (PC)
- B. Data Register (DR)
- C. Instruction Register (IR)
- D. Memory Address Register (MAR)

الإجابة : C

14- The Accumulator Register (AR) is mainly used for:

- A. Storing arithmetic and logical operation results
- B. Holding the address of the next instruction
- C. Managing memory allocation
- D. Controlling the flow of execution

الإجابة : A

15- Which CPU register is responsible for keeping track of the next instruction to be executed?

- A. Accumulator Register (AR)
- B. Instruction Register (IR)
- C. Data Register (DR)
- D. Program Counter (PC)

الإجابة : D

16- What is the function of the Memory Address Register (MAR)?

- A. Holding the address of data or instructions in memory
- B. Performing arithmetic operations
- C. Managing input and output devices
- D. Storing the next instruction to execute

الإجابة : A

17- Which register stores data temporarily before passing it to the ALU for processing?

- A. Instruction Register
- B. Data Register
- C. Program Counter
- D. Cache Memory

الإجابة : B

18- What is an Integrated Circuit (IC)?

- A. A single large transistor
- B. A chip that combines many transistors into one unit
- C. A memory storage device
- D. A power supply unit

الإجابة : B

19- VLSI (Very Large Scale Integration) refers to a chip that contains:

- A. A single transistor
- B. Dozens of transistors
- C. Hundreds of transistors
- D. Hundreds of thousands to millions of transistors

الإجابة : D

20- Why is silicon widely used in electronics?

- A. It is the best conductor in the world
- B. It is a semiconductor that can be modified to conduct or insulate electricity
- C. It is stronger than metal
- D. It is cheaper than plastic

الإجابة : B

21- Which of the following best describes a semiconductor?

- A. A material that is always a conductor
- B. A material that is always an insulator
- C. A material that can act as both a conductor and an insulator under specific conditions
- D. A material that blocks all electricity

الإجابة : C

22- Which of the following best describes a microprocessor?

- A. A processor built into a machine for a specific task
- B. A processor that processes digital signals like audio and video
- C. A processor that handles general computing tasks in a computer
- D. A processor that controls small electronic devices

الإجابة : C

23- What is a characteristic of RISC processors?

- A. Complex and slow instructions
- B. Simple and fast instructions
- C. Uses only 8-bit memory
- D. Designed only for gaming systems

الإجابة : B

24- As you move closer to the CPU in the memory hierarchy, what happens to size and speed?

- A. Size increases, speed decreases
- B. Size decreases, speed increases
- C. Both size and speed increase
- D. Both size and speed decrease

الإجابة : B

25- What was the impact of MOS technology on integrated circuits?

- A. Increased power consumption
- B. Decreased power consumption
- C. Reduced processing speed
- D. Increased transistor size

الإجابة : B

26- What are the key factors affecting a program's performance?

- A. Algorithm, Programming Language, Compiler, Instruction Set Architecture, and Hardware
- B. Operating System, Memory, Network Speed, and Disk Space
- C. Screen Resolution, Battery Life, and Keyboard Type
- D. Internet Connection and Cloud Storage

الإجابة : A

27- How does an algorithm affect the performance of a program?

- A. It only affects the number of instructions executed
- B. It affects instruction count and CPI (Cycles Per Instruction)
- C. It has no impact on performance
- D. It only affects the choice of programming language

الإجابة : B

28- If an algorithm frequently uses division operations, what will happen?

- A. The program will execute faster
- B. The program will have a higher CPI because division takes more cycles
- C. The compiler will remove all divisions automatically
- D. The number of instructions will decrease

الإجابة : B

29- How does a programming language affect performance?

- A. It only affects how readable the code is
- B. It affects instruction count and possibly CPI
- C. It determines the clock rate of the CPU
- D. It has no effect on performance

الإجابة : B

30- Which aspect of performance does the instruction set architecture (ISA) affect?

- A. Instruction count, clock rate, and CPI
- B. Only the clock rate
- C. Only the instruction count
- D. Only the CPI

الإجابة : A

31- If a CPU has 2 billion clock cycles and a clock rate of 1 GHz, what is the CPU execution time?

- A. 1 second
- B. 2 seconds
- C. 0.5 seconds
- D. 4 seconds

الإجابة : B

32- Which of the following best defines execution time?

- A. The time spent by the CPU processing instructions
- B. The time taken for disk access
- C. The time required to perform I/O operations
- D. The time taken by the operating system to switch tasks

الإجابة : A

33- If a program executes 5,000,000 instructions and takes 2 seconds to run, what is the MIPS rating?

- A. 2.5 MIPS
- B. 5 MIPS
- C. 10 MIPS
- D. 250 MIPS

الإجابة : A

34- What is the principle of locality?

- A. Programs access a large portion of their address space at any time.
- B. Programs access a small portion of their address space at any time.
- C. Programs access random portions of their address space.
- D. Programs access only the stack and heap.

الإجابة : B

35- Which of the following is an example of temporal locality?

- A. Accessing elements of an array sequentially.
- B. Accessing data from different memory locations randomly.
- C. Accessing instructions in a loop.
- D. Accessing data from the disk frequently.

الإجابة : C

36- What is the main difference between SRAM and DRAM?

- A. SRAM is faster but more expensive than DRAM.
- B. SRAM is slower but cheaper than DRAM.
- C. SRAM is non-volatile, while DRAM is volatile
- D. SRAM is used for main memory, while DRAM is used for cache.

الإجابة : A

37- What is the miss penalty in cache memory?

- A. The time taken to access data from the cache.
- B. The time taken to replace a block in the cache and deliver it to the processor.
- C. The time taken to write data back to memory.
- D. The time taken to refresh the cache.

الإجابة : B

38- Which of the following is true about a direct-mapped cache?

- A. Each set contains multiple blocks.
- B. Data can be placed in any block in the cache.
- C. It uses a fully associative mapping.
- D. Each set contains exactly one block.

الإجابة : D

39- What is the purpose of a valid bit in a cache?

- A. To indicate whether the data in the cache is dirty.
- B. To indicate whether the data in the cache is valid or not.
- C. To indicate the size of the cache block.
- D. To indicate the number of cache hits.

الإجابة : B

40- What is the formula for Average Memory Access Time (AMAT)?

- A. $AMAT = \text{Hit time} + \text{Miss rate} * \text{Miss penalty}$
- B. $AMAT = \text{Hit time} * \text{Miss rate} + \text{Miss penalty}$
- C. $AMAT = \text{Hit time} / \text{Miss rate} * \text{Miss penalty}$
- D. $AMAT = \text{Hit time} - \text{Miss rate} * \text{Miss penalty}$

الإجابة : A

41- Which of the following is a characteristic of a fully associative cache?

- A. Each block can be placed in any cache entry.
- B. Each block is mapped to a specific set.
- C. It uses a direct-mapped approach.
- D. It has a fixed number of blocks per set.

الإجابة : A

42- What is the main advantage of using a write-back cache policy?

- A. It writes data to memory immediately.
- B. It reduces the number of writes to memory by delaying them.
- C. It increases the cache hit rate.
- D. It reduces the cache size.

الإجابة : B

43- Temporal locality refers to the tendency of a program to access data that is near recently accessed data.

- A. True
- B. False

(This is spatial locality)

الإجابة : B

44- DRAM requires periodic refreshing because it uses capacitors to store data.

A. True

B. False

الإجابة : A

45- The hit rate of a cache is calculated as the number of misses divided by the total number of accesses.

A. True

B. False

(Hit rate is hits/accesses)

الإجابة : B

46- In a n_way associative cache, a block can be placed in any cache entry.

A. True

B. False

الإجابة : B

47- The miss penalty is the time taken to access data from the cache.

A. True

B. False

(Miss penalty is the time to replace a block and deliver it to the processor)

الإجابة : B

48- SRAM is typically used for main memory due to its low cost and high density.

A. True

B. False

(DRAM is used for main memory)

الإجابة : B

49- In a write-through cache, data is written to both the cache and main memory simultaneously.

- A. True
- B. False

الإجابة : A

50- The block size in a cache is typically measured in bits.

- A. True
- B. False

(Block size is typically measured in bytes)

الإجابة : B

51- The tag in a cache is used to identify which block of memory is stored in a cache location.

- A. True
- B. False

الإجابة : A

52- What is the primary purpose of virtual memory?

- A. To speed up CPU operations
- B. To allow multiple applications to run simultaneously
- C. To extend the amount of usable memory beyond physical RAMa
- D. To maintain system security

الإجابة : C

53- Which of the following is NOT an advantage of using virtual machines?

- A. Improved resource utilization
- B. Increased security through isolation
- C. High physical hardware costs
- D. Ability to run incompatible operating systems

الإجابة : C

54- What component is responsible for translating virtual addresses to physical addresses?

- A. CPU
- B. Disk drive
- C. Page table
- D. Cache memory

الإجابة : C

55- What happens when a page fault occurs?

- A. The system runs out of memory
- B. A program attempts to access a page not in RAM
- C. A virtual machine crashes
- D. All applications close

الإجابة : B

56- Which of the following is a characteristic of a Virtual Machine Monitor (VMM)?

- A. It runs only one operating system
- B. It provides direct hardware access to VMs
- C. It manages multiple virtual machines on a host
- D. It requires more physical memory than the applications themselves

الإجابة : C

57- What role does a Translation Lookaside Buffer (TLB) serve in a computer system?

- A. Storing user data
- B. Caching recent virtual-to-physical address translations
- C. Managing file storage
- D. Increasing CPU clock speed

الإجابة : B

58- The virtual operating system supports a set of instructions related to virtual operating systems.

- A. True
- B. False

الإجابة : A

59- The page table is used to determine the location of a page in memory.

- A. True
- B. False

الإجابة : A

60- When an interrupt occurs, the operating system determines the next process to execute.

- A. True
- B. False

الإجابة : A

61- Virtual memory uses only physical memory and does not include any disk storage

A. True

B. False

الإجابة : B

62- Handling a page fault can be managed either by hardware or software.

A. True

B. False

الإجابة : A

63- What is the primary function of the Memory Address Register (MAR)?

A. Holds the data to be written to memory

B. Specifies the address of the instruction to be read from or written to memory

C. Holds the next instruction to be executed

D. Stores the result of arithmetic operations

الإجابة : B

64- During the fetch cycle, which register holds the address of the next instruction to be fetched?

A. Memory Buffer Register (MBR)

B. Instruction Register (IR)

C. Program Counter (PC)

D. Accumulator

الإجابة : C

65- What is the purpose of the Memory Buffer Register (MBR)?

- A. It holds the address of the next instruction
- B. It holds the data read from or to be written to memory
- C. It stores the opcode of the current instruction
- D. It increments the program counter

الإجابة : B

66- In the fetch sequence, what happens during time unit t1?

- A. $MBR \leftarrow (memory)$
- B. $MAR \leftarrow (PC)$
- C. $PC \leftarrow (PC) + 1$
- D. $IR \leftarrow (MBR)$

الإجابة : B

67- Which of the following is true about micro-operations?

- A. They are complex operations that take multiple clock cycles to complete
- B. They are simple, atomic operations performed by the CPU
- C. They are only used during the execute cycle
- D. They are used to handle interrupts

الإجابة : B

68- What is the role of the Instruction Register (IR)?

- A. It holds the address of the next instruction
- B. It holds the data to be written to memory
- C. It contains the most recently fetched instruction
- D. It increments the program counter

الإجابة : C

69- During the indirect cycle, what is loaded into the MAR?

- A. The address from the Program Counter (PC)
- B. The address from the Instruction Register (IR)
- C. The data from the Memory Buffer Register (MBR)
- D. The result of an arithmetic operation

الإجابة : B

70- What happens during the interrupt cycle?

- A. The contents of the PC are saved to memory
- B. The contents of the IR are saved to memory
- C. The contents of the MAR are saved to memory
- D. The contents of the MBR are saved to memory

الإجابة : A

71- Which of the following is NOT a type of micro-operation?

- A. Transfer data between registers
- B. Transfer data from registers to external memory
- C. Perform arithmetic or logical operations
- D. Execute complex software routines

الإجابة : D

72- What is the function of the Control Unit in a CPU?

- A. To perform arithmetic and logical operations
- B. To manage the flow of data between registers and memory
- C. To generate control signals for executing micro-operations
- D. To store the results of computations

الإجابة : C

73- Which of the following is a characteristic of hardwired control units?

- A. They are flexible and easy to modify
- B. They use microcode to generate control signals
- C. They are difficult to design and test
- D. They are slower than microprogrammed control units

الإجابة : C

74- What is the purpose of the clock in a CPU?

- A. To synchronize the execution of micro-operations
- B. To store the current instruction
- C. To hold the address of the next instruction
- D. To perform arithmetic operations

الإجابة : A

75- Which register is used to hold the opcode of the current instruction?

- A. Program Counter (PC)
- B. Memory Buffer Register (MBR)
- C. Instruction Register (IR)
- D. Memory Address Register (MAR)

الإجابة : C

76- What is the main disadvantage of hardwired control units?

- A. They are too slow
- B. They are difficult to modify and add new instructions
- C. They require more power
- D. They are less reliable

الإجابة : B

77- During the execute cycle of an ADD instruction, what happens in time unit t_3 ?

- A. $MAR \leftarrow (IR \text{ address})$
- B. $MBR \leftarrow (\text{memory})$
- C. $R1 \leftarrow R1 + (MBR)$
- D. $PC \leftarrow (PC) + 1$

الإجابة : C

78- What is the purpose of the Instruction Cycle Code (ICC)?

- A. To hold the address of the next instruction
- B. To indicate the current phase of the instruction cycle (fetch, indirect, execute, interrupt)
- C. To store the result of arithmetic operations
- D. To manage interrupts

الإجابة : B

79- Which of the following is NOT a basic element of a processor?

- A. ALU
- B. Registers
- C. Control Unit
- D. Hard Disk

الإجابة : D

80- What is the function of the ALU in a CPU?

- A. To manage the flow of data between registers
- B. To perform arithmetic and logical operations
- C. To generate control signals
- D. To store the current instruction

الإجابة : B

81- Which of the following is a control signal generated by the Control Unit?

- A. Data to be written to memory
- B. Address of the next instruction
- C. Signal to read from memory
- D. Result of an arithmetic operation

الإجابة : C

82- What is the purpose of the Program Counter (PC)?

- A. To hold the current instruction
- B. To hold the address of the next instruction to be executed
- C. To store data read from memory
- D. To perform arithmetic operations

الإجابة : B

83- In micro-programmed approach, the signals are generated by:

- A. System programs
- B. Utility tools
- C. Machine instructions
- D. None of them

الإجابة : C

84- The reason for the implementation of the cache memory is:

- A. To increase the internal memory of the system
- B. The difference in speeds of operation of the processor and memory
- C. To reduce the memory access and cycle time
- D. All of the mentioned

الإجابة : B,C

الإجابة B صحيحة (بحسب الموقع) لأن (Cache Memory) هي ذاكرة صغيرة وسريعة تستخدم لتقليل الفجوة بين سرعة المعالج و سرعة الذاكرة الرئيسية (RAM) و الإجابة C (لتقليل وقت الوصول إلى الذاكرة وزمن الدورة) هي نتيجة لسبب رئيسي وهو الفرق في السرعة بين المعالج والذاكرة لكن في سلم التصحيح الإجابة المحددة هي C.

85- The temporal aspect of the locality of reference means:

- A. That the recently executed instruction won't be executed soon
- B. That the recently executed instruction is temporarily not referenced
- C. That the recently executed instruction will be executed soon again
- D. None of the above

الإجابة : C

86- The spatial aspect of the locality of reference means:

- A. That the recently executed instruction is executed again next
- B. That the recently executed instruction won't be executed again
- C. That the instruction executed will be executed at a later time
- D. That the instruction in close proximity of the instruction executed will be executed in the future

الإجابة : D

87- The commitment unit uses a queue called:

- A. Record buffer
- B. Commitment buffer
- C. Storage buffer
- D. None of them

الإجابة : A

88- What is the purpose of micro-instruction executions?

- A. Generate data result
- B. Generate a control signal to compile
- C. Generate a control signal to execute
- D. All of the mentioned

الإجابة : C

89- The performance of scaler processors as compared to super-scaler processors is:

- A. Faster
- B. Slower
- C. Similar
- D. None of the above

الإجابة : B

90- Problems with hard wired designs:

- A. Complex sequencing and micro-operation logic
- B. Difficult to design and test
- C. Inflexible design and difficult to add new instructions
- D. All of these

الإجابة : D

91- The page number is used as part of a virtual address to identify pages

- A. True
- B. False

الإجابة : A

92- Where is the TLB stored?

- A. Ram
- B. Cache
- C. Hard disk
- D. None of the above

الإجابة : B

93- Direct mapping will have more misses than set associative mapping

- A. True
- B. False

الإجابة : A

94- In one method of the write policies each cache line has a dirty bit to:

- A. Indicate if cache line has been modified or not
- B. Indicate if RAM line has been modified or not
- C. Indicate if cache line and RAM have been modified or not
- D. None of the above

الإجابة : A

95- The page table to access disk is used to resolve page faults

- A. True
- B. False

الإجابة : A

96- IC combined dozens to hundreds of transistors into more chips

- A. True
- B. False

الإجابة : A

97- Clock circuit converts the analog pulse generated by crystal oscillator to a steady stream of digital clock pulses represented by square wave

A. True

B. False

الإجابة : A

98- In sRAM: data is stored as a charge in a capacitor

A. True

B. False

الإجابة : B

99- Capacity miss occurs when the set of active cache blocks is larger than the cache

A. True

B. False

الإجابة : A

100- Instruction set: تربط بين العتاد الصلب والبرمجيات ولها العديد من الأشكال (ARM - RISC - MIPS) وتختلف من حاسوب لآخر

A. True

B. False

الإجابة : A

100- إن جميع العمليات الصغيرة تقع في إحدى الفئات التالية:

- A. نقل المعطيات من سجل إلى آخر
- B. نقل المعكيات من سجل إلى واجهة ربط خارجية مثل مسرى النظام
- C. نقل المعطيات من واجهة ربط خارجية إلى سجل
- D. إجراء عملية منطقية أو حسابية باستخدام سجلات الدخل والخرج
- E. A و B و C

الإجابة : E

101- Physical address determines the size of:

- A. Virtual memory
- B. Physical memory
- C. RAM
- D. ROM
- E. Cache memory

الإجابة : B

102- Which of the following memory types is used for main memory?

- A. ROM and SDRAM
- B. SRAM and DRAM
- C. SDRAM and DRAM
- D. DRAM and EPROM
- E. EPROM and DRAM

الإجابة : C

103- _____ is the fastest type of memory

- A. Cache memory
- B. Register
- C. Main memory
- D. Hard disk

الإجابة : B

104- An optimizing compiler does:

- A. Better memory management
- B. Better compilation of the given piece of code
- C. Taked advantage of the type of processor and reduces its process time

الإجابة : B

105- The control unit controls other units by generating:

- A. Transfer signals
- B. Command signals
- C. Timing signals
- D. None of them

الإجابة : C

106- A sequence of control words corresponding to a control sequence is called:

- A. Micro function
- B. Micro routine
- C. Micro procedure
- D. None of them

الإجابة : B

- A computer has 32 Kbytes of virtual memory and 8 Kbytes of main memory with a page size of 512 bytes:

1- How many bits are in the virtual address?

- A. 15
- B. 10
- C. 20
- D. 12

الإجابة : A

2- How many pages are in the virtual address?

- A. 40
- B. 64
- C. 60
- D. 128

الإجابة : B



"The only way to do great work is to love what you do."

3- How many bits are required for the physical address?

- A. 10
- B. 12
- C. 13
- D. 16

الإجابة : C

نتذكر أن

كيلو $K : 2^{10}$

ميغا $M : 2^{20}$

غيغا $G : 2^{30}$

Virtual memory = 32 Kbytes = $2^5 \times 2^{10}$ bytes = 2^{15} bytes

Main memory size = physical memory size = 8 Kbytes = 2^{13} bytes

Page size = 512 bytes = 2^9 bytes

لدينا:

$$VA = \log_2(\text{virtual memory}) = \log_2(2^{15}) = 15$$

$$PA = \log_2(\text{physical memory}) = \log_2(2^{13}) = 13$$

$$V.\text{pages} = \frac{2^{VA}}{\text{page size}} = \frac{2^{15}}{2^9} = 2^6 = 64$$

4- Consider a cache with 64 blocks and a block size of 16, what block number does byte address 1200 map to?

- A. 13
- B. 10
- C. 11
- D. 12

الإجابة : C

$$\text{Block number} = \left(\frac{\text{Byte address}}{\text{Block size}} \right) \bmod (\text{Number of the blocks})$$

← باقي قسمة

ولدينا من السؤال:

64 block = Blocks عدد الـ

16 bytes = Block size

وعنوان البايت المطلوب (1200)

$$\text{Block number} = \left(\frac{1200}{16} \right) \bmod (64) = 11$$

■ Assume for a given machine and program:

Instruction cache miss rate is 2% and data cache miss rate is 4% and miss penalty is always 40 cycles, CPI of 2 without memory stalls and frequency of load/stores is 36% of the instructions

5- How much faster is a machine with a perfect cache that never misses?

- A. 1.688
- B. 1.912
- C. 2
- D. 1.5

الإجابة : A

6- What happens if we speed up the machine by reducing its CPI to 1 without changing the clock rate?

- A. 1.516
- B. 2
- C. 2.376
- D. 2.992

الإجابة : C

7- If we speed up the machine by doubling its clock rate, but the absolute time for a miss penalty remains the same:

Performance with a perfect cache is better by a factor of:

- A. 2.376
- B. 2.72
- C. 3.762
- D. 4

الإجابة : A

من النص لدينا:

- Instruction cache miss rate = 2%
- Data cache miss rate = 4%
- Miss penalty = 40 cycles
- Frequency of load/store = 36% of the instructions

السؤال الأول:

سنحسب الـ misses التي توجد ضمن البرنامج ونضيف قيمتهم لقيمة الـ CPI بدون إيقافات:

$$\text{Instruction misses} = 0.02 \times 40 = 0.8 \text{ cpi}$$

$$\text{Data misses} = 0.04 \times 40 \times 0.36 = 0.576 \text{ cpi}$$

فالدورات المضافة الكلية للـ CPI هي:

$$0.8 + 0.576 = 1.376 \text{ نجمعها مع CPI بدون إيقافات} \Leftarrow$$

$$\text{CPI} = 2 + 1.376$$

لحساب الأسرع:

$$\frac{\text{CPI}_{\text{new}}}{\text{CPI}_{\text{old}}} = \frac{3.376}{2} = 1.688$$

نسبة السرعة بين معالج بدون misses مع معالج مع misses.

السؤال الثاني:

$$1 + 1.376 = 2.376$$

قيمة الـ misses من السؤال السابق

فتكون قيمة سرعة الأداء:

$$\frac{2.376}{1} = 2.376$$

السؤال الثالث:

مضاعفة قيمة الـ clock rate تؤثر على الـ penalty فتصبح:

$$40 \times 2 = 80$$

نحسب الآن:

$$\text{Instruction misses} = 0.02 \times 80 = 1.6 \text{ cpi}$$

$$\text{Data misses} = 0.36 \times 0.04 \times 80 = 1.152$$

$$\text{Total cpi} = 2 + 1.152 + 1.6 = 4.752$$

فيكون مقدار السرعة الفارقة أو speed up:

$$\frac{4.752}{2} = 2.376$$

8- The directly mapped cache no replacement algorithm is required.

E. True

F. False

الإجابة : A

9- The surroundings of the recently accessed block is called as _____

E. Neighborhood

F. Neighbor

G. Locality of reference

H. None of the mentioned

الإجابة : C

10- In set associative and associative mapping there exists less flexibility.

E. True

F. False

الإجابة : B

11-The algorithm which replaces the block which has not been referenced for a while is called

E. LRU

F. ORF

G. Direct

H. Both LRU and ORF

الإجابة : A

12- In associative mapping during LRU, the counter of the new block is set to '0' and all the others are incremented by one when _____ occurs.

- E. Delay
- F. Miss
- G. Hit
- H. Delayed hit

الإجابة : B

13- The LRU provides very bad performance when it comes to _____

- E. Blocks being accessed is sequential
- F. When the blocks are randomized
- G. When the consecutive blocks accessed are in the extremes
- H. None of the mentioned

الإجابة : A

14- The algorithm which removes the recently used page first is _____

- E. LRU
- F. MRU
- G. OFM
- H. None of the mentioned

الإجابة : B

15- The LRU can be improved by providing a little randomness in the access.

- E. True
- F. False

الإجابة : A



16- In LRU, the referenced blocks counter is set to '0' and that of the previous blocks are incremented by one and others remain same, in the case of _____

- E. Hit
- F. Miss
- G. Delay
- H. None of the mentioned

الإجابة : A

17- The counter that keeps track of how many times a block is most likely used is _____

- E. Count
- F. Reference counter
- G. Use counter
- H. Probable count

الإجابة : B

18- A word whose individual bits represent a control signal is _____

- E. Command word
- F. Control word
- G. Co-ordination word
- H. Generation word

الإجابة : B



19- Individual control words of the micro routine are called as _____

- E. Micro task
- F. Micro operation
- G. Micro instruction
- H. Micro command

الإجابة : C

20- The special memory used to store the micro routines of a computer is _____

- E. Control table
- F. Control store
- G. Control mart
- H. Control shop

الإجابة : B

21- Every time a new instruction is loaded into IR the output of _____ is loaded into UPC

- E. Starting address generator
- F. Loader
- G. Linker
- H. Clock

الإجابة : A



22- The effectiveness of the cache memory is based on the property of _____

- E. Locality of reference
- F. Memory localization
- G. Memory size
- H. None of the mentioned

الإجابة : A

23- The algorithm to remove and place new contents into the cache is called _____

- E. Replacement algorithm
- F. Renewal algorithm
- G. Updation
- H. None of the mentioned

الإجابة : A

24- The copy-back protocol is used _____

- E. To copy the contents of the memory onto the cache
- F. To update the contents of the memory from the cache
- G. To remove the contents of the cache and push it on to the memory
- H. None of the mentioned

الإجابة : B

25- The bit used to signify that the cache location is updated is _____

- E. Dirty bit
- F. Update bit
- G. Reference bit
- H. Flag bit

الإجابة : A

قسم الدكتور سيرا

1- What is the primary goal of pipelining in processors?

- A. Reduce latency of a single instruction
- B. Increase throughput by overlapping instruction execution
- C. Simplify the data path design
- D. Eliminate all structural hazards

الإجابة : B

2- In the laundry analogy for pipelining, what determines the throughput?

- A. The slowest subtask (e.g., washer)
- B. The number of loads divided by the total time
- C. The number of subtasks (n) in the pipeline
- D. The time to complete one load

الإجابة : C

3- Why is MIPS architecture well-suited for pipelining?

- A. Variable-length instructions
- B. Symmetric instruction formats and fixed instruction length
- C. Limited memory access instructions
- D. Complex control logic

الإجابة : B

4- In a pipelined MIPS data path, which stage calculates the memory address for a load/store instruction?

- A. Instruction Fetch (IF)
- B. Instruction Decode (ID)
- C. Execute (EX)
- D. Memory Access (MEM)

الإجابة : C

5- A single-cycle processor has a clock cycle time of 8 ns. For three instructions, the total execution time is:

- A. 8 ns
- B. 16 ns
- C. 24 ns
- D. 32 ns

$$\text{CpuTime} = ic * cpi * \text{clock cycle time} = 3 * 1 * 8 = 24 \text{ ns}$$

الإجابة : C

6- In a 5-stage pipelined processor with a 2 ns clock cycle, how much time is needed to complete three instructions?

- A. 6 ns
- B. 10 ns
- C. 14 ns
- D. 18 ns

$$\text{Cpu Time} = (3 * 1 + 4) * 2 = 14 \text{ ns}$$

الإجابة : C

-الشرح :

$$cc = Pl.depth + (ic - 1) + stalls = 5 + 2 + 0 = 7 \quad ; \quad CPUtime = cc * T = 7 * 2 = 14 \text{ ns}$$

-بطريقة أخرى :

دورات التعبئة $\text{Cycle drain} = n - 1$

n هي عدد المراحل المقسمة لها العملية و بعد دورات التعبئة مع كل نبضة تنفذ عملية

$$CPUtime = (ic * cpi + cycle drain) * clock cycle time = ((3 * 1) + 4) * 2 = 14 \text{ ns}$$

7- Which pipeline register holds the sign-extended immediate value and source registers?

- A. IF/ID
- B. ID/EX
- C. EX/MEM
- D. MEM/WB

الإجابة : B



8- Consider 5-stages pipelined Datapath. What happened in 2nd stage for load (w) instruction:

- A. The instruction being read from memory using the address in the PC and then being placed in the IF/ID pipeline register.
The PC address is incremented by 4 and then written back into the PC to be ready for the next clock cycle.
This incremented address is also saved in the IF/ID pipeline register
- B. The instruction portion of the IF/ID pipeline register supplying the 16-bit immediate field, Which is sign-extended to 32bits, and the register numbers to read the two registers.
All three values are stored in the ID/EX pipeline register, along with the incremented PC address
- C. The load instruction reads the contents of register 1 and sign-extended immediate from the id/ex pipeline register and add them using the ALU. That sum is placed in the EX/MEM pipeline register
- D. The load instruction reading the data memory using the address from the EX/MEM pipeline register and loading the data into the MEM/WB pipeline register.

الإجابة : B

9- Consider the following code order in pipelined Datapath:

lw \$s0, 20(\$t1)

sub \$t2, \$s0, \$t3

How to solve the hazard in the previews code?

- A. Using stall operation
- B. Code reordering
- C. Using Forwarding
- D. Adding multiple function units

الإجابة : A+C

10- Structural hazards occur due to:

- A. Data dependencies between instructions
- B. Resource conflicts (e.g., shared memory)
- C. Incorrect branch predictions
- D. Unequal pipeline stage lengths

الإجابة : B

11- Which solution forwards results from the EX/MEM register to the ALU to resolve data hazards?

- A. Stalling
- B. Bypassing
- C. Code reordering
- D. Branch prediction

الإجابة : B

12- What to do if data hazard detected?

- A. wait/stall
- B. Reorder in Software (SW)
- C. Forward/Bypass
- D. All the above
- E. None. we will use some other method.

الإجابة : D

13- Pipelining makes it possible for processor to execute instructions in parallel:

- A. True
- B. False

الإجابة : A

14- Pipeline Hazard is: situation in pipelining
when the next instruction cannot execute in the next clock cycle?

- A. True
- B. False

الإجابة : A

15- Pipelining: is an implementation technique in which multiple instructions
are overlapped in execution?

- A. True
- B. False

الإجابة : A

16- In MIPS instruction execution of the pipelined processor,

The WB stage is the :

- A. first stage
- B. third stage
- C. fourth stage
- D. fifth stage

الإجابة : D

17- occurs when an instruction depends on the result of the previous instruction in a way that is exposed by the overlapping of an instructions in the pipeline.

- A. structural hazards
- B. data hazards
- C. control hazards
- D. Hazard in the pipeline

الإجابة : B

18- Any condition that causes a processor to stall is called as _____

- A. Hazard
- B. Page fault
- C. System error
- D. None of the mentioned

الإجابة : A

19- The stalling of the processor due to the unavailability of the instructions is called as :

- A. control Hazard
- B. structural Hazard
- C. input Hazard
- D. None of the mentioned

الإجابة : A

20- What is Static Branch Prediction?

- A. A solution to Control Hazard, rely on the processor to predict branches at run time
- B. A solution to Control Hazard, by using Branch Prediction Buffer
- C. A solution to Control Hazard, rely on the compiler to predict branches at compile time
- D. A solution to Control Hazard, by using Branch History table

الإجابة : C

21- In a 1-bit branch prediction buffer, if a branch is mispredicted:

- A. The history bit is inverted immediately
- B. The buffer is flushed
- C. Two mispredictions are required to change the bit
- D. The pipeline stalls

الإجابة : A

22- A 2-bit branch predictor improves accuracy by:

- A. Requiring two mispredictions to change the prediction state
- B. Using static prediction
- C. Eliminating all branch hazards
- D. Doubling the buffer size

الإجابة : A

23- The contention for the usage of a hardware device is called _____

- A. Structural hazard
- B. Stalk
- C. Deadlock
- D. None of the mentioned

الإجابة : A

الشرح:

contention : نزاع

24- The time lost due to the branch instruction is often referred to as _____

- A. Latency
- B. Delay
- C. Branch penalty
- D. None of the mentioned

الإجابة : C

25- The pipeline bubbling is a method used to prevent data hazard and structural hazards.

- A. True
- B. False

الإجابة : A

26- an instruction j is data dependent on instruction I if either of the following holds:

- A. When the two instruction I and j use same register or memory location
- B. Instruction j is data dependent on instruction k, and instruction k is data dependent on instruction i
- C. Instruction I produce a result that may be used by instruction j
- D. When the two instructions I and j contain Branch or Jump.

الإجابة : C,B



27- Consider the following code:

```

Loop: L.D F0, 0(R1)
      ADD .D F4, F0, F2
      S.D F4, 0(R1 )
      DADDUI R1,R1,#-8
      BNE R1, R2, Loop
  
```

What kinds of Dependencies we have in the previous code?

- A. 4 data dependencies and 1 name dependency
- B. 4 data dependencies and 4 name dependencies
- C. 3 data dependencies and 2 name dependencies
- D. 3 data dependencies and 1 name dependency

الإجابة : D

الشرح:

■ Data dependencies:

بين التعليمة الأولى التي تكتب على F0 و الثانية التي تقرأ منه
 بين التعليمة الثانية و الثالثة حيث الثانية تكتب على F4 و الثالثة تقرأ منه
 بين التعليمة الرابعة و الخامسة حيث الرابعة تكتب على R1 و الثانية تقرأ منه

■ Name dependency:

بين التعليمة الثالثة و الرابعة حيث الثالثة تقرأ من R1 و الرابعة تكتب عليه و يسمى هذا النوع ب Anti_dependency

28- Data Hazard exists whenever there is a name or data dependence between close enough instruction, and May be classified as:

- A. RAW
- B. WAW
- C. WAR
- D. All of the above

الإجابة : D

ملاحظة:

RAR isn't a hazard

29- What are the constraints imposed control dependencies:

- A. An instruction that is not control dependent on a branch cannot be moved before the branch so that its execution will be controlled by the branch.
- B. An instruction that is control dependent on a branch cannot be moved after the branch because its execution will not be controlled by the branch
- C. An instruction that is control dependent on a branch cannot be moved before the branch because its execution will not be controlled by the branch
- D. An instruction that is not control dependent on a branch cannot be moved after the branch so that its execution will be controlled by the branch

الإجابة : D,C

ملاحظة:

قيود مفروضة: Constraints imposed

30- Output dependence occurs when instruction j writes a register or memory location that instruction i reads?

- A. True
- B. False

الإجابة : B

الشرح:

output dependency هو الكتابة على نفس السجل في كلا التعليمتين مع المحافظة على ترتيب التعليمات و لحل هذه التبعية يمكننا إعادة تسمية السجلات.

31- _____ Is a standard technique for removing false data dependencies. _____ & _____ dependencies , among register data.

- A. Register naming , WAR,WAW
- B. WAR,WAW , shelving
- C. Register naming , shelving
- D. None

الإجابة : A

32- To discard instructions in the pipeline is referred to as :

- A. Decoding
- B. Discarding
- C. Flushing
- D. Load_use

الإجابة : C

الشرح:

- Freeze the pipeline holding any instruction after the branch instruction (Stall)
- Flush the pipeline to delete the instructions after the branch if condition is evaluated, and branch is to take

33- What is the main goal of increasing the pipeline depth in processors?

- A. Reducing the number of instructions that can be executed simultaneously
- B. Increasing the likelihood of hazards
- C. Improving performance by increasing instruction-level parallelism (ILP)
- D. Reducing the complexity of processor design

الإجابة : C

34- What are the two main methods to increase Instruction-Level Parallelism (ILP)?

- A. Increasing pipeline depth and using speculation
- B. Increasing pipeline depth and using multiple issue
- C. Using speculation and reordering instructions
- D. Increasing pipeline depth and reducing the number of executed instructions

الإجابة : B



35- What does "Static Multiple Issue" mean in processors?

- A. Issuing multiple instructions at runtime based on processor analysis
- B. Issuing multiple instructions randomly without prior analysis
- C. Issuing multiple instructions at compile time based on compiler analysis
- D. Issuing multiple instructions only in case of hazards

الإجابة : C

36- Which policy allows issuing instructions in a specific order but completing them in a different order?

- A. In-Order Issue with In-Order Completion
- B. Out-of-Order Issue with Out-of-Order Completion
- C. Static Issue with Dynamic Completion
- D. In-Order Issue with Out-of-Order Completion

الإجابة : D

37- What is speculation in processors?

- A. Executing instructions before confirming their correctness based on prediction
- B. Delaying instruction execution until hazards are confirmed
- C. Reordering instructions to avoid hazards
- D. Halting the processor until instruction correctness is confirmed

الإجابة : A

38- What is one of the main problems that can occur due to speculation?

- A. Increasing processor speed
- B. The possibility of executing incorrect instructions
- C. Reducing the number of hazards
- D. Improving memory performance

الإجابة : B

39- What is the goal of using "Loop Unrolling"?

- A. Reducing the number of instructions in the program
- B. Increasing the number of loop iterations
- C. Improving performance by reducing hazards and increasing parallelism
- D. Reducing the number of registers used

الإجابة : C

40- What is one of the techniques used to avoid hazards in pipelining?

- A. Increasing the number of pipeline stages
- B. Using speculation
- C. Register renaming
- D. Reducing the number of executed instructions

الإجابة : C

41- What is "Dynamic Branch Prediction"?

- A. Predicting the instruction path at compile time
- B. Predicting the instruction path at runtime
- C. Delaying instruction execution until the correct path is confirmed
- D. Halting the processor until the correct path is determined

الإجابة : B

42- What is one of the problems that can occur when using "Loop Unrolling"?

- A. Increasing the number of instructions in the program
- B. Reducing the number of registers used
- C. Reducing the number of hazards
- D. Improving memory performance

الإجابة : A



43- What is the purpose of using a "Reorder Buffer (ROB)" in processors?

- A. Storing instructions that were executed incorrectly
- B. Storing temporary results of instructions until their correctness is confirmed
- C. Reducing the number of hazards
- D. Increasing processor speed

الإجابة : B

44- What is one of the techniques used to avoid "Name Dependencies" in loops?

- A. Increasing the number of pipeline stages
- B. Using speculation
- C. Register renaming
- D. Reducing the number of executed instructions

الإجابة : C

45- What is "Memory Disambiguation"?

- A. Determining memory contents to avoid hazards
- B. Increasing memory speed
- C. Reducing the number of memory-related instructions
- D. Halting the processor until memory contents are determined

الإجابة : A

46- What is one of the main benefits of using "Dynamic Scheduling" in processors?

- A. Reducing the number of executed instructions
- B. Increasing parallelism by scheduling instructions at runtime
- C. Reducing the number of hazards
- D. Halting the processor until memory contents are determined

الإجابة : B

47- What is the purpose of using a “Delayed Branch Slot” in processors?

- A. Delaying instruction execution after a branch to avoid executing incorrect instructions
- B. Increasing processor speed
- C. Reducing the number of hazards
- D. Improving memory performance

الإجابة : A

48- Which design method breaks down complex functions?

- A. Top Down
- B. Bottom-Up
- C. Sequential
- D. Parallel

الإجابة : A

49- How many bits are needed to encode 10 states?

- A. 1
- B. 3
- C. 4
- D. 6

الإجابة : C $2^n=10 \Rightarrow n=4$

50- Next-state function is a:

- A. Sequential
- B. Combinational
- C. Parallel
- D. Associative

الإجابة : B

51- Output function produces a set of outputs from :

- A. The current state and the inputs
- B. The current state and the outputs
- C. The next state and the inputs
- D. The next state and the outputs

الإجابة : A

52- Which instruction type uses the funct field in MIPS?

- A. R-type
- B. Load
- C. Store
- D. Branch

الإجابة : A

53- What is the first step in FSM design?

- A. State Diagram
- B. Truth Table
- C. Karnaugh Map
- D. Logic Circuit

الإجابة : A

54- Memory, registers, adders, and ALU are:

- A. Control elements.
- B. State elements.
- C. Datapath elements
- D. Combinational elements

الإجابة : C

55- In Finite State Machine (FSM), Next state function is :

- A. a Combinational function that given the inputs and the current state,
- B. determines the next state of the system.
- C. a sequential function that given the inputs and the current state, determines the next state of the system.
- D. a Combinational function that given inputs and the current state, determines the outputs of the system.
- E. a Function produces a set of outputs from the current state and the inputs

الإجابة : A

56- Sequential system can be described with a truth table ?

- A. Ture
- B. False

الإجابة : B

57- Control Signals are generated by a control unit consisting of one or more Finite State Machines?

- A. Ture
- B. False

الإجابة : A